

MDM2 – Case Study: Assistive Robotics for Improvement of Life Quality One-Page Proposal

Team	5
Members	Vishnu Thaimachedath Nandakumar & Jeethu Thambi
Project Title	Sensor based Rehab Glove
GitHub Repository URL	https://github.com/vishnutnGithub/assistive-robotics-life-quality.git
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Industrial Application (target domain/use-case)	Healthcare / Assistive Robotics / Home Rehabilitation
Keywords (3–6)	Haptic gloves, hand rehabilitation, assistive robotics, wearable sensing, finger assistance, home therapy
Submission Date (YYYY-MM-DD)	10/04/2026
Gantt Chart	Gantt Chart – Haptic Glove Project Timeline

1) Problem Statement & Measurable Outcomes (3–4 sentences)	Hand injuries, stroke, and neurological disorders often reduce finger mobility and grip strength, making rehabilitation slow and repetitive. Traditional therapy requires frequent clinic visits, which can be expensive and inconvenient. This project proposes a wearable haptic glove that assists finger movement and supports repetitive rehabilitation exercises at home.
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	Measurable outcomes (KPIs): • Improve finger extension angle by at least 20% • Increase grip repetition count per session • Reduce glove weight below 500 g for comfort • Track therapy accuracy through sensor feedback																		
2) Motivation & Assistive Relevance (2–3 sentences)	Patients recovering from stroke or hand injury benefit from continuous guided rehabilitation, especially outside hospitals. A wearable glove provides affordable daily therapy support, increases exercise consistency, and may improve recovery speed. This is important because home-based rehabilitation demand is growing rapidly.																		
3) Method & Feasibility (≤6 sentences)	The project will review existing rehabilitation glove mechanisms and compare soft actuator and cable-driven designs. Sensor options such as flex sensors will be evaluated for finger tracking. A conceptual glove model will be designed emphasizing lightweight wearable comfort. Basic actuation logic and therapy feedback methods will be proposed. Expected outputs include design sketches, component selection, and functional workflow diagrams. A literature-based comparison table will support feasibility.																		
4) Milestones & Timeline (short table/list)	<table><tr><th>Phase</th><th>Task</th><th>Target</th></tr><tr><td>P1</td><td>Topic selection + proposal</td><td>Week 1</td></tr><tr><td>P2</td><td>Literature review</td><td>Week 2</td></tr><tr><td>P3</td><td>Concept design</td><td>Week 3</td></tr><tr><td>P4</td><td>Sensor + actuator analysis</td><td>Week 4</td></tr><tr><td>P5</td><td>Final presentation/report</td><td>Week 5</td></tr></table>	Phase	Task	Target	P1	Topic selection + proposal	Week 1	P2	Literature review	Week 2	P3	Concept design	Week 3	P4	Sensor + actuator analysis	Week 4	P5	Final presentation/report	Week 5
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5) Risks & Ethics (1–2 sentences)	Patient rehabilitation data must remain private if digital tracking is used. Design recommendations should avoid unrealistic medical claims and properly cite all technical sources.																		

BILL OF MATERIALS

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PROJECT NAME	HAPTIC GLOVES FOR HAND REHABILITATION		
ITEMS	DESCRIPTION	QTY	COST
Servo Motors	MG90S, 5V- 6V	3	€ 15.00
Flex Sensor		3	€ 60.00
Force Sensor	FSR 402	3	€ 40.00
Microcontroller	Teensy 4.1	1	€ 40.00
Vibration Motor		3	€ 6.00
Battery	Lithium Ion, 11V	1	€ 20.00
Voltage Regulator	3-5 V	3	€ 12.00
TOTAL	€ 193.00		